

2025

DESCON TWINKAMPUS

Mentors:

Pushpender

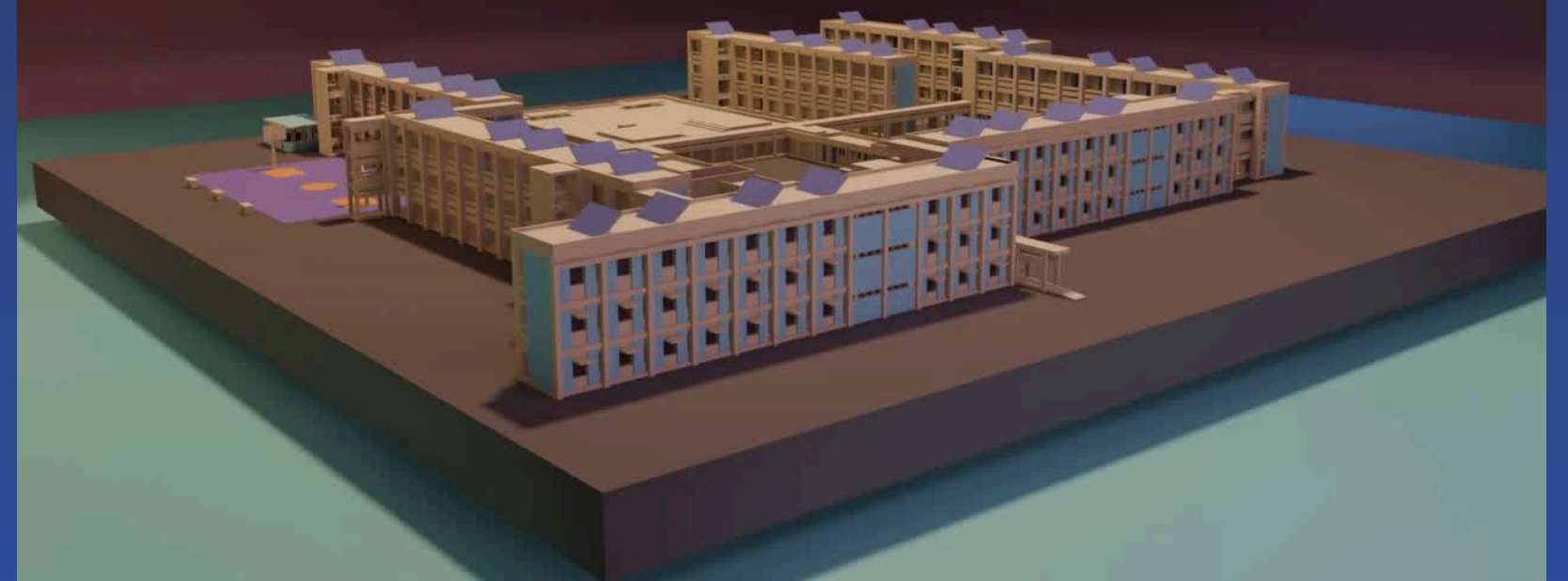
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Project Overview

This project focused on creating a digital twin—a simple 3D version—of campus buildings using tools like AutoCAD, Fusion 360, Blender, and SolidWorks. We built the 3D models using floor plans and satellite images, then exported them in web formats like .glb. These models were brought to life on a React-based website with Three.js, allowing users to explore the campus interactively in their browser. The project was completed individually with student-friendly tools and helped build practical skills in 3D modeling and web development.

HALL OF RESDIENCE 3



OBJECTIVES



Why this project?

- Offers an interactive 3D view of campus buildings, replacing static layouts.
- Helps users—students, staff, and visitors—understand the campus better.
- Requires no institutional access; can be completed independently.
- Combines engineering design with modern web technologies.



Objectives

- Design simplified 3D models of campus buildings using tools like AutoCAD, Fusion 360, Blender, and SolidWorks.
- Convert those models into web-compatible formats for online viewing.
- Build an interactive website using React and Three.js to explore models in real-time.
- Strengthen practical skills in 3D modeling, web development, and visualization.

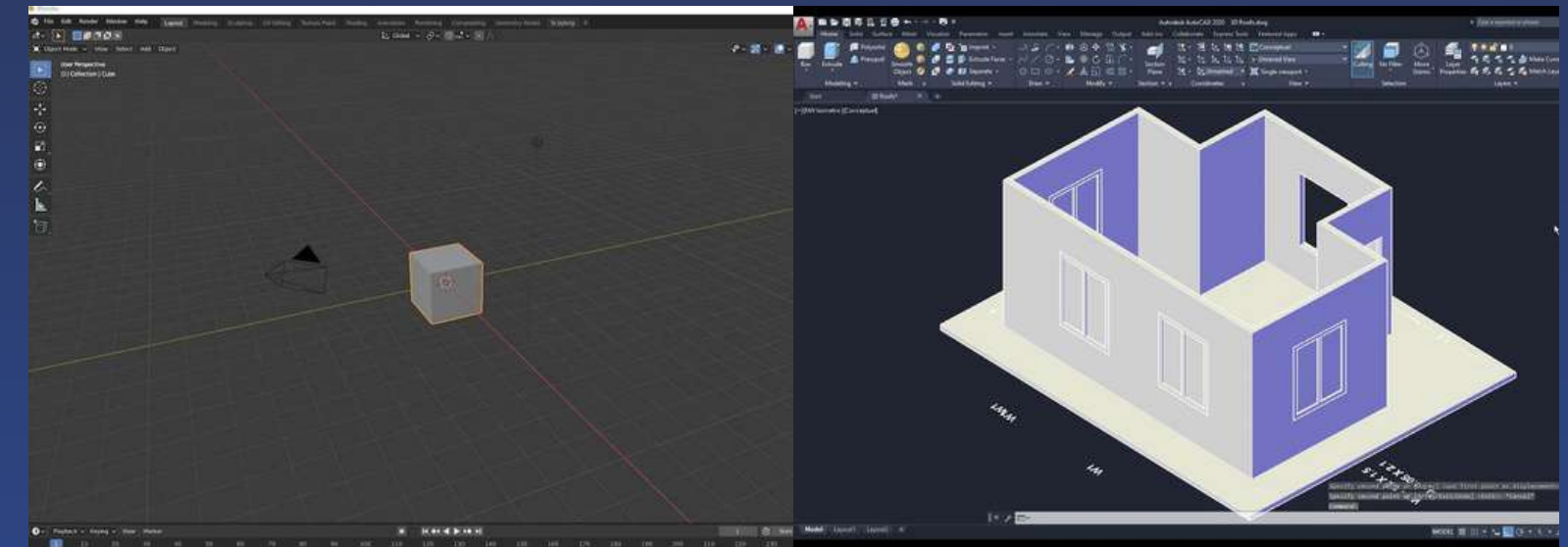
AutoCAD

AutoCAD is a computer-aided design (CAD) software developed by Autodesk, used for creating precise 2D drawings and 3D models. It's widely used by architects, engineers, and designers to draft building plans, mechanical parts, and infrastructure layouts.



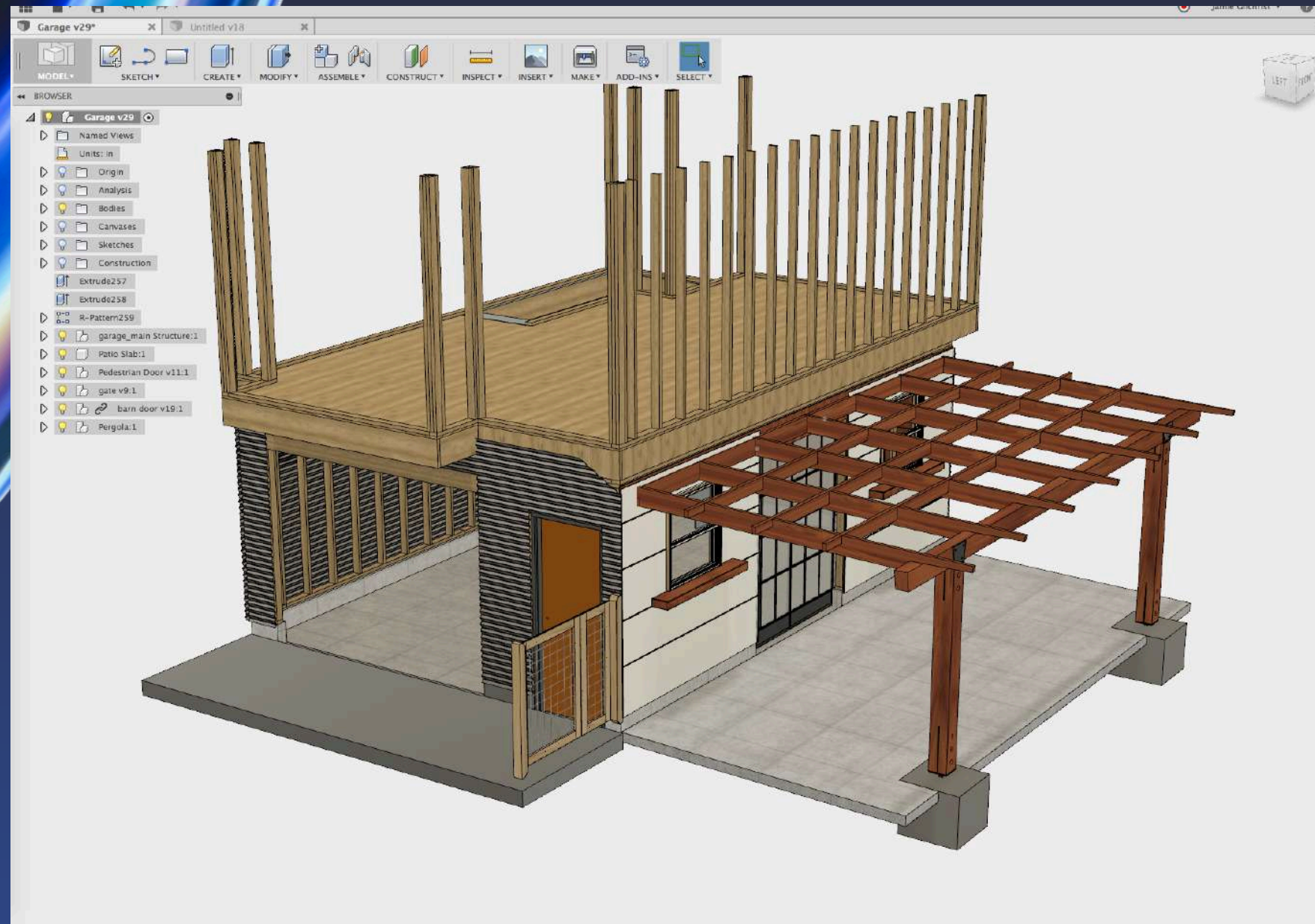
Blender

Blender is a free and open-source 3D creation suite used for modeling, animation, rendering, and video editing. It's popular among artists, designers, and developers for creating everything from animated films and game assets to visual effects and 3D-printed models.

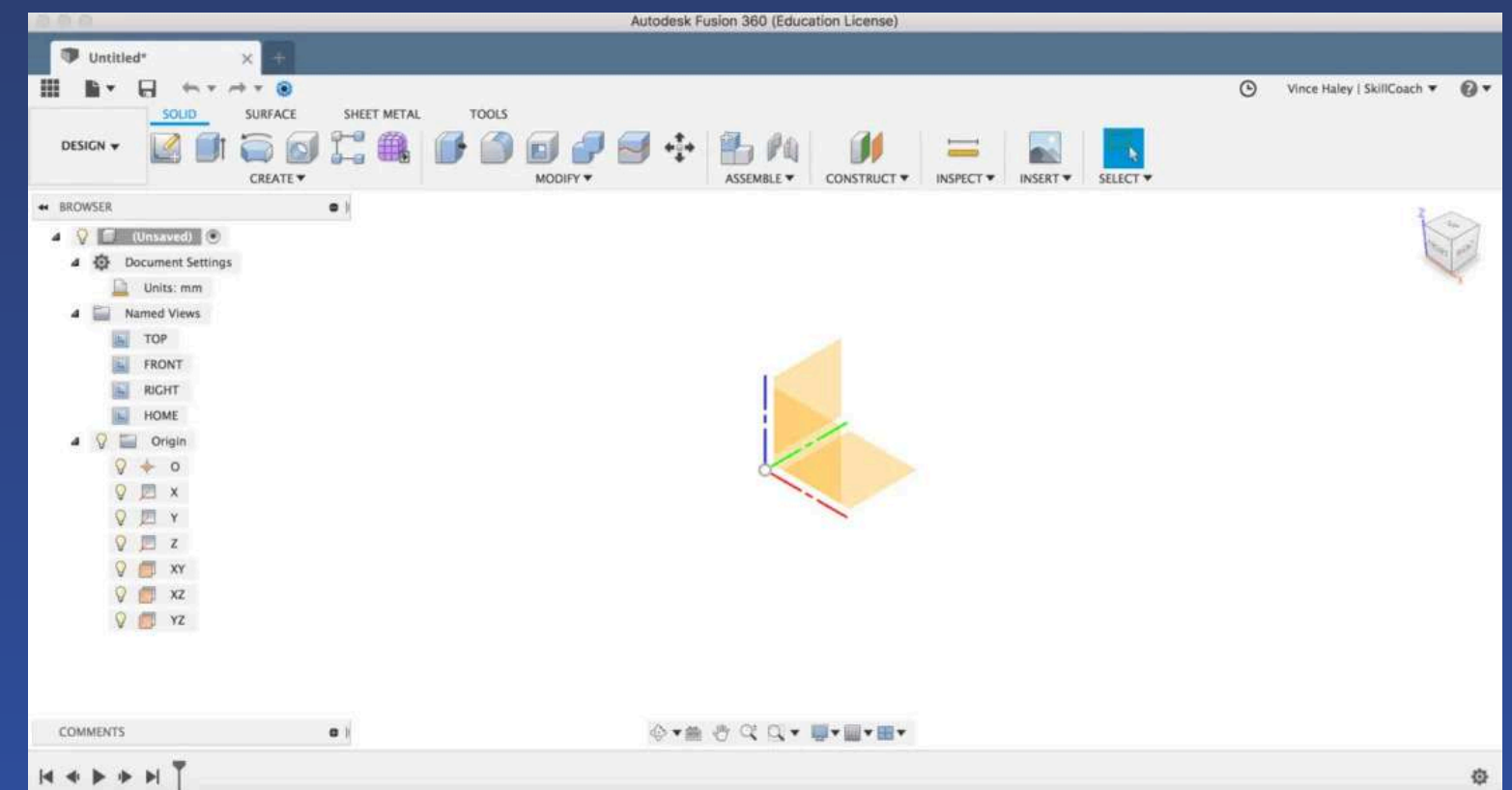


SOFTWARES

FUSION 360



Fusion 360 is a versatile 3D design and engineering tool used for creating, simulating, and refining digital models. It brings together modeling, simulation, and manufacturing features in one workspace, making it ideal for product development and mechanical design.



WORKFLOW

3D Modelling
(AutoCAD, Fusion etc.)

Exporting Format
(FBX, GLB etc.)

Hosting
(Vercel, GitHub etc.)

Integration into Web
(THREE.js, React)



MODELING PROCESS



WEB INTEGRATION

```
JS main.js  X  # index.css
JS main.js > loadModel
43 function loadModel() {
48   (glTF) => {
53     object.traverse((child) => {
58       child.material.roughness = 0.9;
59       if (child.material.color) {
60         child.material.color.multiplyScalar(0.98);
61       }
62       child.material.envMapIntensity = 1.0;
63       child.material.needsUpdate = true;
64     });
65   });
66
67   scene.add(object);
68
69   // Compute bounding box
70   const box = new THREE.Box3().setFromObject(object);
71   const center = box.getCenter(new THREE.Vector3());
```

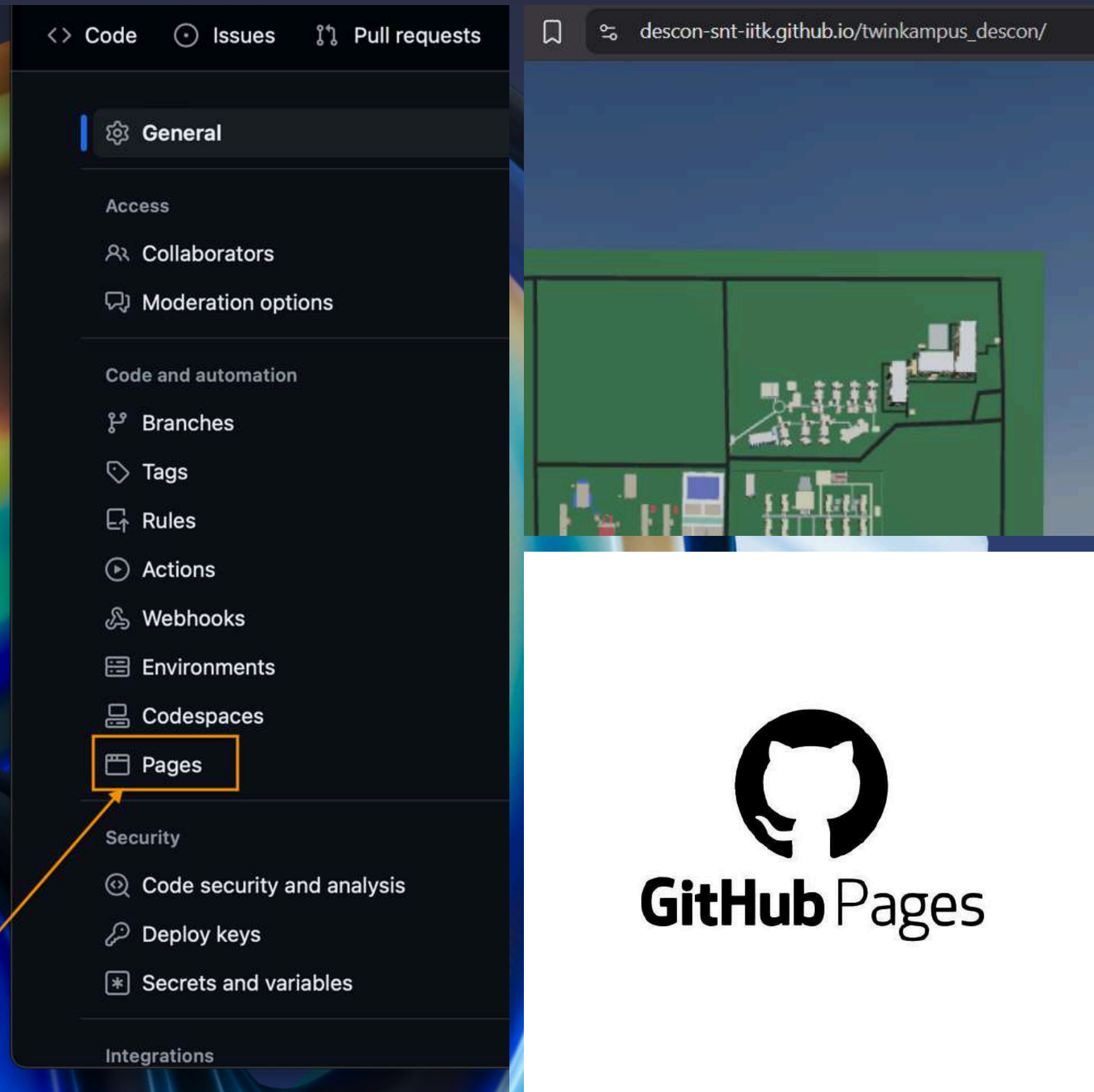
```
JS main.js  X  # index.css
JS main.js > loadModel
1  import * as THREE from "three";
2  import { OrbitControls } from "three/examples/jsm/controls/OrbitControls.js";
3  import { RGBELoader } from "three/examples/jsm/loaders/RGBELoader.js";
4  import { GLTFLoader } from "three/examples/jsm/loaders/GLTFLoader.js";
5
6  // Scene, Camera, Renderer
7  const scene = new THREE.Scene();
8
9  const camera = new THREE.PerspectiveCamera(
10    60,
11    window.innerWidth / window.innerHeight,
12    0.1,
13    1000
14  );
15  camera.position.set(0, 1, 3);
16
17  const renderer = new THREE.WebGLRenderer({ antialias: true });
18  renderer.setSize(window.innerWidth, window.innerHeight);
19  renderer.toneMapping = THREE.ACESFilmicToneMapping;
20  renderer.toneMappingExposure = 0.8; // reduce brightness
```

```
JS main.js  # index.css  X
# index.css > html
1  @import "tailwindcss";
2
3  html,
4  body {
5    margin: 0;
6    padding: 0;
7    overflow: hidden;
8    height: 100%;
9    background-color: #000;
10 }
11
12 canvas {
13   display: block;
14   position: absolute;
15   top: 0;
16   left: 0;
17 }
18
```

```
JS main.js  X  # index.css
JS main.js > loadModel
96 // Balanced lighting
97 const hemiLight = new THREE.HemisphereLight(0xffffff, 0x444444, 0.3);
98 scene.add(hemiLight);
99
100 const dirLight = new THREE.DirectionalLight(0xffffff, 0.5);
101 dirLight.position.set(5, 10, 7.5);
102 scene.add(dirLight);
103
104 // Responsive resize
105 window.addEventListener("resize", () => {
106   camera.aspect = window.innerWidth / window.innerHeight;
107   camera.updateProjectionMatrix();
108   renderer.setSize(window.innerWidth, window.innerHeight);
109 });
110
111 // Animation loop
112 function render() {
113   requestAnimationFrame(render);
114   controls.update();
115   renderer.render(scene, camera);
116 }
117 render();
118
```

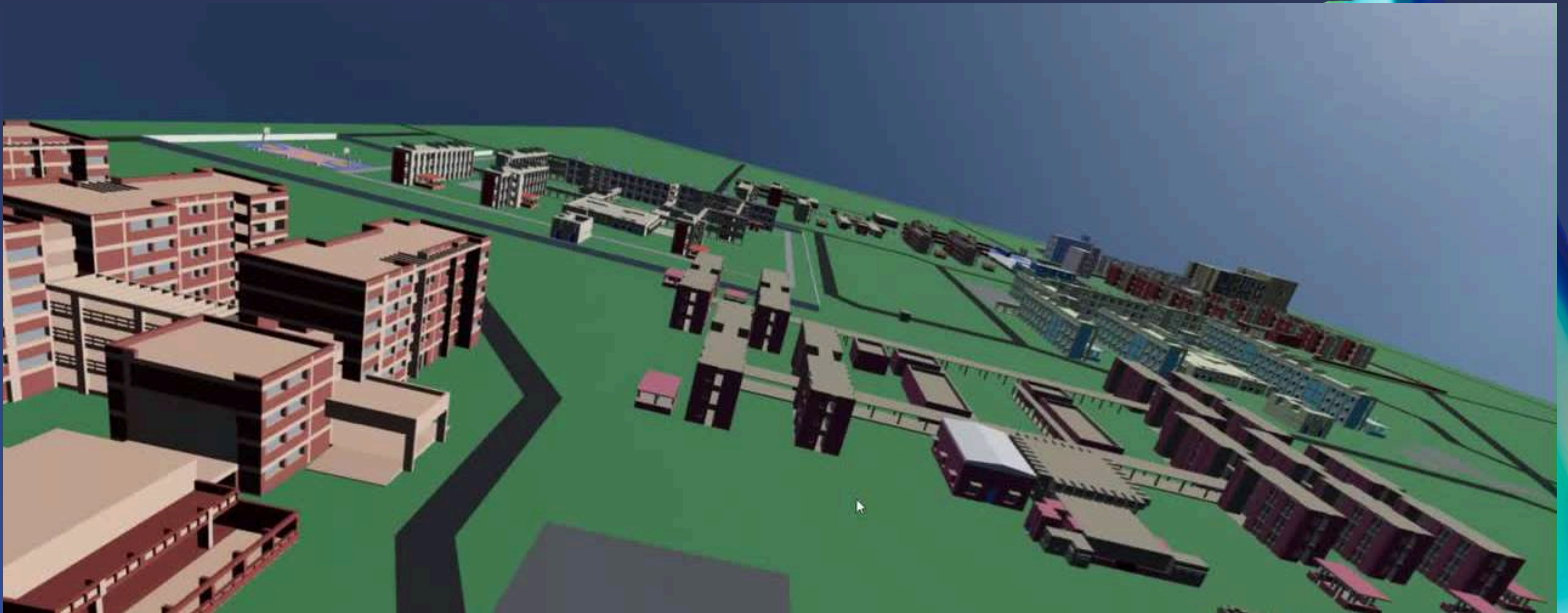

DEPLOYMENT

For this project, we used GitHub Pages to deploy our web-based viewer, making it accessible to anyone with a browser. After building the React + Three.js application, we pushed the code to a GitHub repository and configured GitHub Pages to serve the site directly from the repository. This method is free, reliable, and perfect for hosting static websites like ours. It allowed us to showcase the 3D campus models online without needing any external servers or paid hosting services.



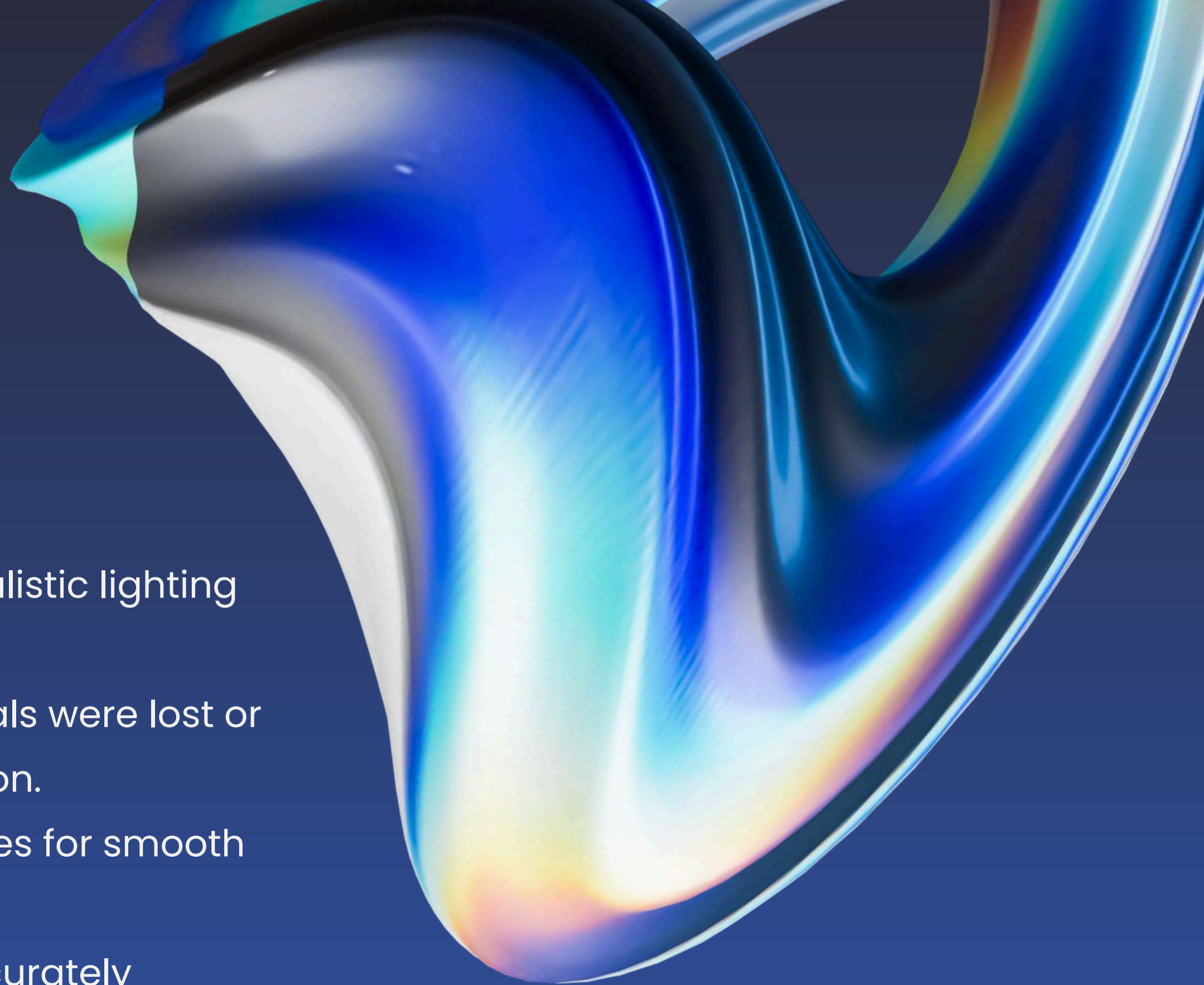
FINAL RESULT

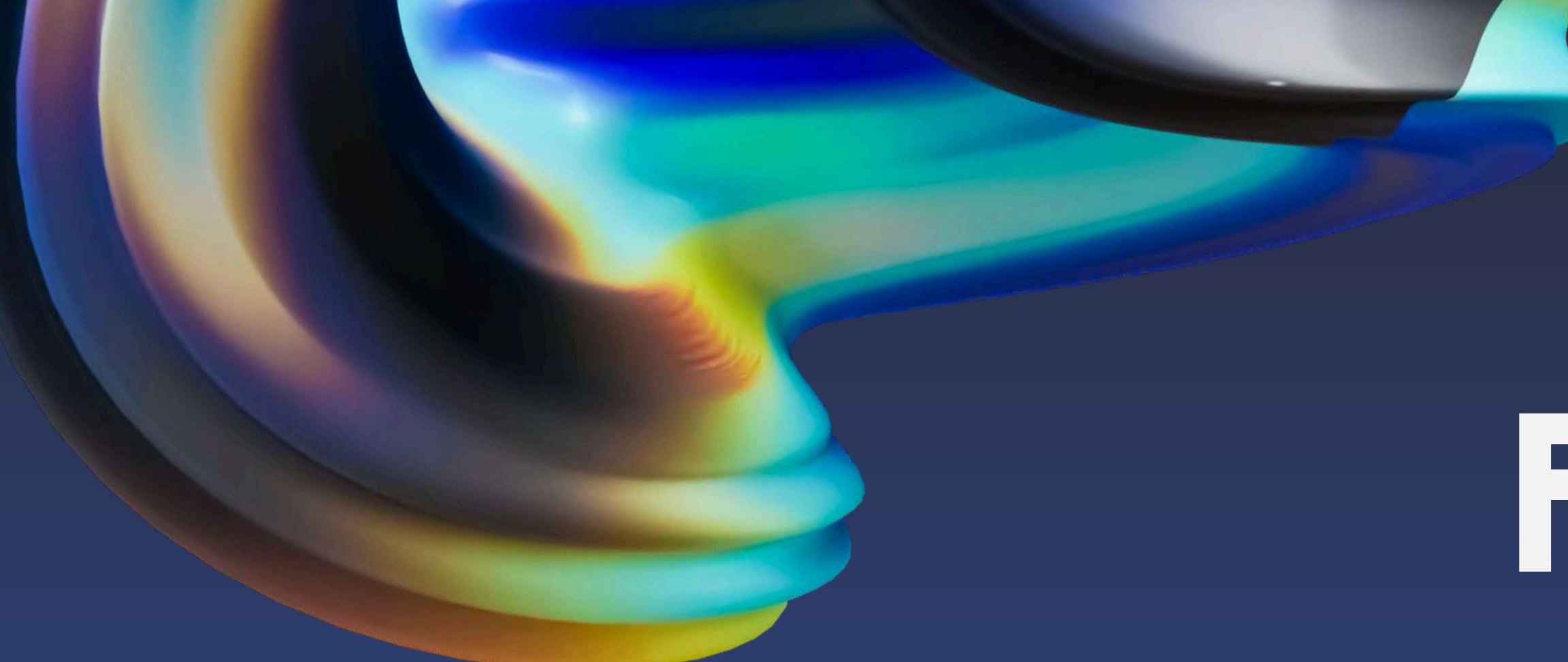
https://descon-snt-iitk.github.io/twinkampus_descon/



CHALLENGES FACED

- **Lighting Issues in Three.js:** Achieving realistic lighting effects.
- **Loss of Materials on Conversion:** Materials were lost or not properly transferred during conversion.
- **File Size Optimization:** Optimizing file sizes for smooth loading and rendering.
- **Texture Mapping Issues:** Difficulty in accurately mapping textures onto 3D models.





LEARNINGS & FUTURE GOALS

- **Technical Learnings:**

- Mastery of 3D formats
- Understanding of WebGL
- Familiarity with hosting web application

- **Personal Learnings:**

- Effective debugging techniques
- Importance of thorough documentation
- Strategies for optimization
- Time management skills

The project has potential for expansion and enhancement. Some future features include:

- **Enhanced User Experience:**

- Mobile responsiveness
- Multi-model switcher

- **Immersive Experiences:**

- AR/VR integration
- Campus-wide 3D tour
- Street View



THANK YOU